

## CHATHAM 1 SE, MI, USA

WMO: 725380

Lat: 46.3345N

Lon: 86.9200W

Elev: 875

StdP: 14.24

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54810

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -10.1      | -4.8 | -17.2                         | 2.2 | -10.3 | -12.1 | 3.0 | -4.6 | 15.3                  | 15.7 | 14.2 | 16.5 | 2.4                   | N/A  | 0.520 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.1                   | 84.2            | 70.3 | 81.1 | 68.5 | 78.2 | 66.4 | 72.3                | 80.2 | 70.3 | 77.9 | 68.4 | 75.5 | 3.8                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.5                            | 112.0 | 76.5 | 67.7 | 105.0 | 74.6 | 65.8 | 98.4 | 72.4 | 36.5          | 80.1 | 34.8 | 78.2 | 33.2 | 75.5 | 80.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 12.4          | 10.7 | 9.2 | DB                         | -19.7 | 89.1               | 7.6 | 2.9                                                | -25.2 | 91.2       | -29.6 | 92.9       | -33.9 | 94.5       | -39.4 | 96.6 | (4) |
| (5)               |      |     | WB                         | -21.6 | 76.6               | 8.2 | 2.4                                                | -27.5 | 78.3       | -32.3 | 79.7       | -36.9 | 81.1       | -42.8 | 82.8 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 41.0   | 16.8  | 16.1  | 26.0  | 37.6  | 50.0  | 59.1  | 64.8  | 63.3  | 56.9  | 44.9  | 32.9  | 21.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 19.53  | 10.73 | 11.11 | 11.40 | 8.88  | 8.90  | 7.15  | 6.42  | 5.90  | 7.87  | 8.45  | 9.34  | 9.73 | (7)  |
| (8)  |                                                                                  | HDD50       | 4866   | 1031  | 949   | 749   | 383   | 113   | 10    | 0     | 0     | 23    | 211   | 517   | 880  | (8)  |
| (9)  |                                                                                  | HDD65       | 8959   | 1496  | 1369  | 1210  | 821   | 472   | 203   | 85    | 104   | 264   | 626   | 964   | 1345 | (9)  |
| (10) |                                                                                  | CDD50       | 1567   | 0     | 0     | 4     | 12    | 113   | 284   | 458   | 412   | 229   | 52    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD65       | 185    | 0     | 0     | 0     | 0     | 7     | 27    | 79    | 50    | 20    | 2     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 1787   | 0     | 0     | 2     | 2     | 103   | 290   | 781   | 453   | 146   | 10    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 365    | 0     | 0     | 0     | 0     | 16    | 52    | 193   | 80    | 24    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.9    | 4.9   | 4.9   | 4.7   | 4.8   | 4.1   | 3.1   | 2.8   | 2.7   | 2.9   | 3.6   | 4.2   | 4.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 31.9   | 1.8   | 1.3   | 1.8   | 2.0   | 2.8   | 3.0   | 3.2   | 3.3   | 4.1   | 3.6   | 3.1   | 2.0  | (15) |
| (16) |                                                                                  | PrecMax     | 41.7   | 5.2   | 4.2   | 4.7   | 4.4   | 7.0   | 6.7   | 6.9   | 8.0   | 7.5   | 8.0   | 7.4   | 5.1  | (16) |
| (17) |                                                                                  | PrecMin     | 22.4   | 0.4   | 0.1   | 0.4   | 0.4   | 0.2   | 0.6   | 0.7   | 0.4   | 0.9   | 1.1   | 0.2   | 0.4  | (17) |
| (18) |                                                                                  | PrecStd     | 5.4    | 1.0   | 0.7   | 1.0   | 1.0   | 1.5   | 1.5   | 1.7   | 1.5   | 1.6   | 1.6   | 1.4   | 0.9  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 40.3  | 44.0  | 63.6  | 71.1  | 82.1  | 85.0  | 89.0  | 86.4  | 83.8  | 75.0  | 61.4  | 45.0 | (19) |
| (20) |                                                                                  |             | MCWB   | 36.5  | 38.7  | 52.3  | 53.5  | 64.0  | 70.8  | 74.1  | 71.6  | 69.9  | 60.7  | 51.7  | 41.6 | (20) |
| (21) |                                                                                  | 2%          | DB     | 36.7  | 38.7  | 51.8  | 63.7  | 76.9  | 81.0  | 85.1  | 82.1  | 78.0  | 68.0  | 53.5  | 39.6 | (21) |
| (22) |                                                                                  |             | MCWB   | 34.5  | 35.0  | 46.9  | 51.4  | 62.0  | 67.7  | 70.9  | 69.1  | 67.5  | 58.1  | 47.9  | 38.0 | (22) |
| (23) |                                                                                  | 5%          | DB     | 34.2  | 34.1  | 46.5  | 58.4  | 71.6  | 77.2  | 81.8  | 79.0  | 74.3  | 63.4  | 49.7  | 36.8 | (23) |
| (24) |                                                                                  |             | MCWB   | 32.8  | 31.3  | 41.0  | 45.8  | 58.9  | 64.8  | 69.7  | 66.6  | 65.0  | 56.7  | 45.5  | 35.3 | (24) |
| (25) |                                                                                  | 10%         | DB     | 32.0  | 31.7  | 41.5  | 53.2  | 66.6  | 73.9  | 78.3  | 76.1  | 70.8  | 58.6  | 46.1  | 34.4 | (25) |
| (26) |                                                                                  |             | MCWB   | 30.6  | 29.6  | 37.0  | 43.7  | 56.5  | 62.9  | 67.3  | 65.7  | 63.6  | 54.0  | 43.1  | 33.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 37.1  | 41.1  | 59.2  | 55.4  | 68.3  | 73.0  | 76.5  | 73.8  | 72.6  | 63.0  | 53.9  | 44.6 | (27) |
| (28) |                                                                                  |             | MCD B  | 38.6  | 44.9  | 69.6  | 63.3  | 76.8  | 81.8  | 84.7  | 82.3  | 79.8  | 67.9  | 59.0  | 45.8 | (28) |
| (29) |                                                                                  | 2%          | WB     | 34.6  | 35.8  | 48.5  | 50.4  | 65.0  | 69.5  | 73.1  | 71.1  | 69.5  | 59.9  | 49.9  | 38.9 | (29) |
| (30) |                                                                                  |             | MCD B  | 36.1  | 39.2  | 51.8  | 56.9  | 72.1  | 78.0  | 81.0  | 78.5  | 74.8  | 65.3  | 53.4  | 39.9 | (30) |
| (31) |                                                                                  | 5%          | WB     | 32.7  | 32.2  | 41.1  | 45.5  | 61.7  | 66.5  | 70.8  | 68.9  | 67.2  | 57.1  | 45.7  | 35.9 | (31) |
| (32) |                                                                                  |             | MCD B  | 34.0  | 34.3  | 45.0  | 54.2  | 69.2  | 73.3  | 79.0  | 75.9  | 71.9  | 61.5  | 49.0  | 36.9 | (32) |
| (33) |                                                                                  | 10%         | WB     | 30.2  | 29.9  | 36.2  | 42.1  | 58.6  | 64.0  | 68.9  | 66.9  | 64.7  | 53.9  | 42.8  | 33.5 | (33) |
| (34) |                                                                                  |             | MCD B  | 31.5  | 32.0  | 39.5  | 49.7  | 65.3  | 71.1  | 76.6  | 73.1  | 68.9  | 58.1  | 45.7  | 34.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.5  | 17.3  | 18.0  | 18.6  | 21.7  | 22.0  | 22.1  | 21.6  | 20.4  | 16.1  | 13.2  | 12.5 | (35) |
| (36) |                                                                                  |             | MCDBR  | 12.9  | 18.2  | 22.7  | 29.7  | 29.0  | 26.7  | 25.7  | 25.7  | 23.9  | 22.4  | 19.3  | 13.2 | (36) |
| (37) |                                                                                  |             | MCWBR  | 11.8  | 14.6  | 15.2  | 18.7  | 16.6  | 14.7  | 14.0  | 14.7  | 13.1  | 14.8  | 13.4  | 12.0 | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 12.9  | 16.7  | 19.6  | 26.0  | 23.8  | 22.4  | 22.9  | 21.8  | 19.9  | 20.4  | 17.5  | 12.4 | (38) |
| (39) |                                                                                  |             | MCWBR  | 11.7  | 13.7  | 14.1  | 17.6  | 15.4  | 14.0  | 13.8  | 13.1  | 13.1  | 14.7  | 12.8  | 11.4 | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.264  | 0.276 | 0.310 | 0.340 | 0.370 | 0.376 | 0.382 | 0.369 | 0.354 | 0.327 | 0.298 | 0.266 | (40) |      |
| (41) |                                                                                  | taud        | 2.448  | 2.396 | 2.393 | 2.396 | 2.377 | 2.391 | 2.394 | 2.429 | 2.473 | 2.530 | 2.532 | 2.493 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 271    | 288   | 292   | 291   | 284   | 282   | 279   | 279   | 273   | 266   | 255   | 258   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 21     | 27    | 32    | 35    | 37    | 37    | 36    | 34    | 30    | 24    | 20    | 18    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 348    | 620   | 1068  | 1417  | 1734  | 1877  | 1918  | 1624  | 1205  | 716   | 384   | 271   | (44) |      |
| (45) |                                                                                  | RadStd      | 28     | 82    | 99    | 145   | 126   | 144   | 96    | 104   | 83    | 83    | 47    | 23    | (45) |      |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (34 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page